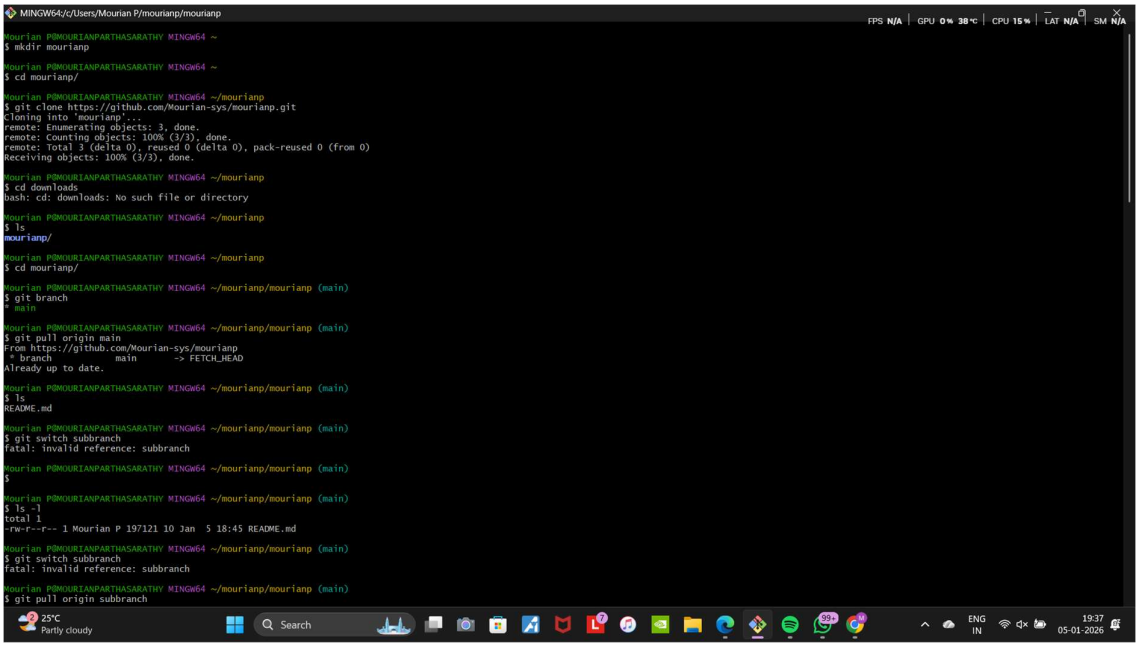
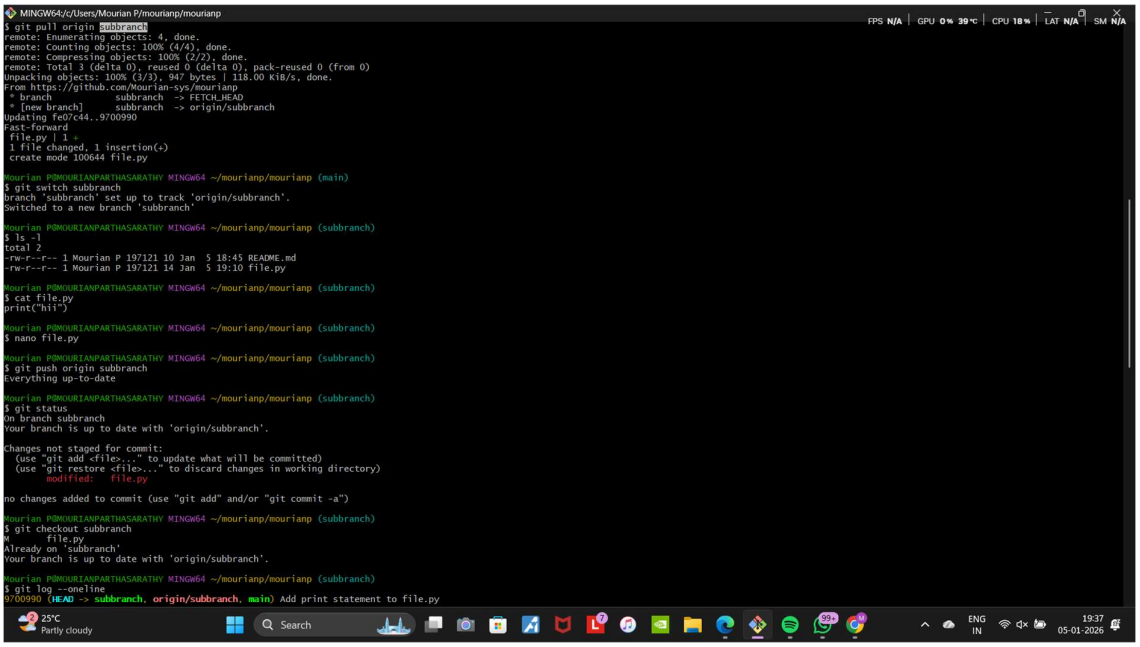
ASSIGNMENT - 8

Git branches , git remote , git pull , git merge , git rebase



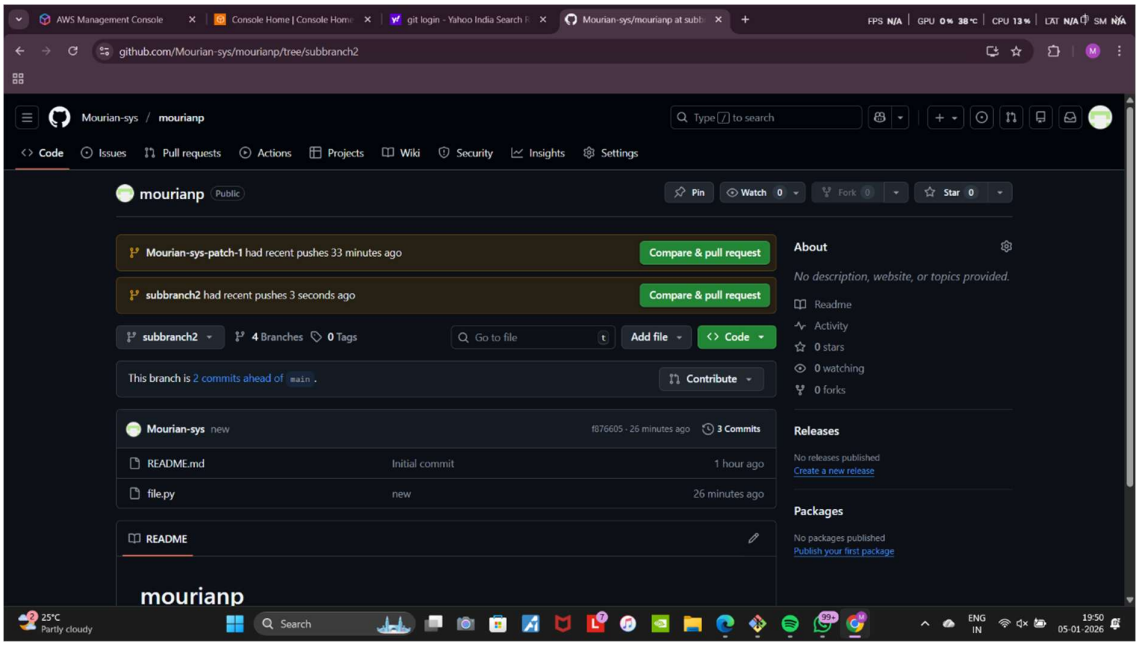
```
MINGW64:/c/Users/Mourian P/mourianp/mourianp
Mourian P@MOURIANPARTHASARATHY MINGW64 ~
$ mkdir mourianp

Mourian P@MOURIANPARTHASARATHY MINGW64 ~
$ cd mourianp/

Mourian P@MOURIANPARTHASARATHY MINGW64 ~/mourianp
$ git clone https://github.com/Mourian-sys/mourianp.git
Cloning into 'mourianp'...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.

Mourian P@MOURIANPARTHASARATHY MINGW64 ~/mourianp
$ cd downloads
bash: cd: downloads: No such file or directory

Mourian P@MOURIANPARTHASARATHY MINGW64 ~/mourianp
$ ls
mourianp/

Mourian P@MOURIANPARTHASARATHY MINGW64 ~/mourianp
$ cd mourianp/

Mourian P@MOURIANPARTHASARATHY MINGW64 ~/mourianp/mourianp (main)
$ git branch
* main

Mourian P@MOURIANPARTHASARATHY MINGW64 ~/mourianp/mourianp (main)
$ git pull origin main
From https://github.com/Mourian-sys/mourianp
 * branch            main       -> FETCH_HEAD
Already up to date.

Mourian P@MOURIANPARTHASARATHY MINGW64 ~/mourianp/mourianp (main)
$ ls
README.md

Mourian P@MOURIANPARTHASARATHY MINGW64 ~/mourianp/mourianp (main)
$ git switch subbranch
fatal: invalid reference: subbranch

Mourian P@MOURIANPARTHASARATHY MINGW64 ~/mourianp/mourianp (main)
$

Mourian P@MOURIANPARTHASARATHY MINGW64 ~/mourianp/mourianp (main)
$ ls -l
total 1
-rw-r--r-- 1 Mourian P 197121 10 Jan  5 18:45 README.md

Mourian P@MOURIANPARTHASARATHY MINGW64 ~/mourianp/mourianp (main)
$ git switch subbranch
fatal: invalid reference: subbranch

Mourian P@MOURIANPARTHASARATHY MINGW64 ~/mourianp/mourianp (main)
$ git pull origin subbranch
```

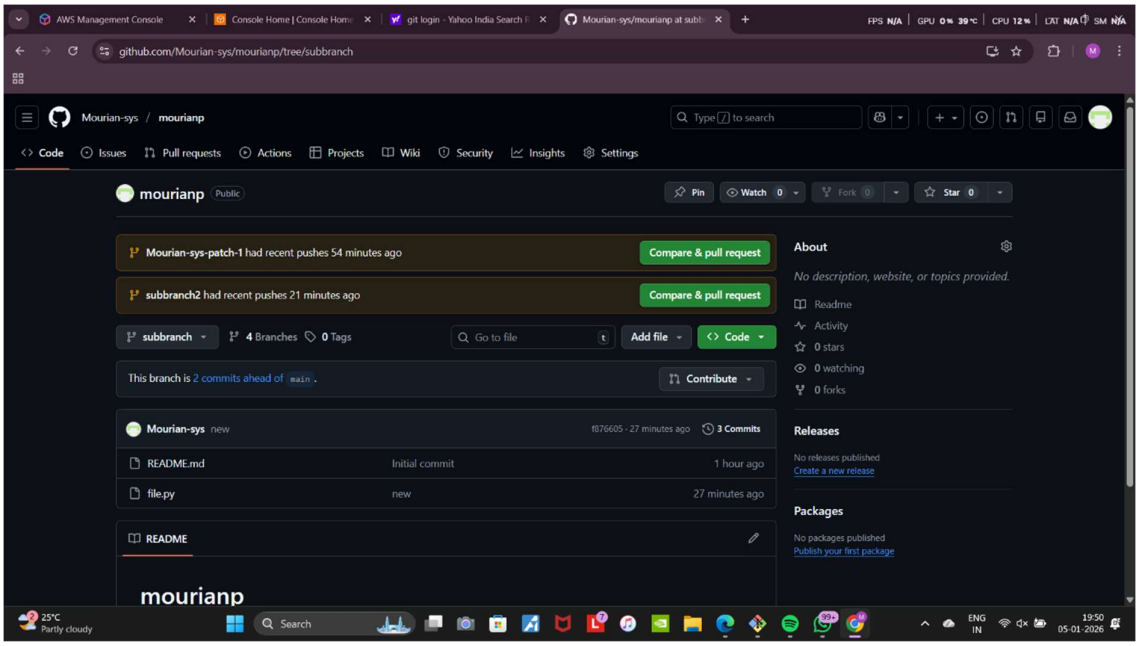


```
MINGW64:/c/Users/Mourian P/mourianp/mourianp
$ git pull origin subbranch
remote: Enumerating objects: 4, done.
remote: Counting objects: 100% (4/4), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 947 bytes | 118.00 KiB/s, done.
From https://github.com/Mourian-sys/mourianp
 * branch            subbranch  -> FETCH_HEAD
 * [new branch]      subbranch  -> origin/subbranch
Updating fe07c44..9700990
Fast-forward
 file.py | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 file.py

Mourian P@MOURIANPARTHASARATHY MINGW64 ~/mourianp/mourianp (main)
$ git switch subbranch
branch 'subbranch' set up to track 'origin/subbranch'.
Switched to a new branch 'subbranch'

Mourian P@MOURIANPARTHASARATHY MINGW64 ~/mourianp/mourianp (subbranch)
$ ls -l
total 2
-rw-r--r-- 1 Mourian P 197121 10 Jan  5 18:45 README.md
-rw-r--r-- 1 Mourian P 197121 14 Jan  5 19:10 file.py

Mourian P@MOURIANPARTHASARATHY MINGW64 ~/mourianp/mourianp (subbranch)
$ cat file.py
print("hii")

Mourian P@MOURIANPARTHASARATHY MINGW64 ~/mourianp/mourianp (subbranch)
$ nano file.py

Mourian P@MOURIANPARTHASARATHY MINGW64 ~/mourianp/mourianp (subbranch)
$ git push origin subbranch
Everything up-to-date

Mourian P@MOURIANPARTHASARATHY MINGW64 ~/mourianp/mourianp (subbranch)
$ git status
On branch subbranch
Your branch is up to date with 'origin/subbranch'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   file.py

no changes added to commit (use "git add" and/or "git commit -a")

Mourian P@MOURIANPARTHASARATHY MINGW64 ~/mourianp/mourianp (subbranch)
$ git checkout subbranch
M       file.py
Already on 'subbranch'
Your branch is up to date with 'origin/subbranch'.

Mourian P@MOURIANPARTHASARATHY MINGW64 ~/mourianp/mourianp (subbranch)
$ git log --oneline
9700990 (HEAD -> subbranch, origin/subbranch, main) Add print statement to file.py
```

```
MINGW64:/c/Users/Mourian P/mourianp/mourianp
9700990 (HEAD -> subbranch, origin/subbranch, main) Add print statement to file.py
fe07c44 (origin/main, origin/HEAD) Initial commit

Mourian P@MOURIANPARTHASARATHY MINGW64 ~/mourianp/mourianp (subbranch)
$ git push origin subbranch
Everything up-to-date

Mourian P@MOURIANPARTHASARATHY MINGW64 ~/mourianp/mourianp (subbranch)
$ cat file.py
print("hji")
print("hi")

Mourian P@MOURIANPARTHASARATHY MINGW64 ~/mourianp/mourianp (subbranch)
$ git add file.py

Mourian P@MOURIANPARTHASARATHY MINGW64 ~/mourianp/mourianp (subbranch)
$ git commit -m "new"
[subbranch f876605] new
 1 file changed, 1 insertion(+)

Mourian P@MOURIANPARTHASARATHY MINGW64 ~/mourianp/mourianp (subbranch)
$ git push origin subbranch
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 16 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 292 bytes | 146.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Mourian-sys/mourianp.git
   9700990..f876605  subbranch -> subbranch

Mourian P@MOURIANPARTHASARATHY MINGW64 ~/mourianp/mourianp (subbranch)
$ git status
On branch subbranch
Your branch is up to date with 'origin/subbranch'.

nothing to commit, working tree clean

Mourian P@MOURIANPARTHASARATHY MINGW64 ~/mourianp/mourianp (subbranch)
$ git pull origin subbranch2
From https://github.com/Mourian-sys/mourianp
 * branch            subbranch2 -> FETCH_HEAD
 * [new branch]      subbranch2 -> origin/subbranch2
Already up to date.

Mourian P@MOURIANPARTHASARATHY MINGW64 ~/mourianp/mourianp (subbranch)
$ git switch subbranch2
branch 'subbranch2' set up to track 'origin/subbranch2'.
Switched to a new branch 'subbranch2'

Mourian P@MOURIANPARTHASARATHY MINGW64 ~/mourianp/mourianp (subbranch2)
$ git pull origin subbranch
From https://github.com/Mourian-sys/mourianp
 * branch            subbranch  -> FETCH_HEAD
Already up to date.

Mourian P@MOURIANPARTHASARATHY MINGW64 ~/mourianp/mourianp (subbranch2)
$ git add subbranch
```

```
MINGW64:/c/Users/Mourian P/mourianp/mourianp
$ git push origin subbranch
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 16 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 292 bytes | 146.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Mourian-sys/mourianp.git
   9700990..f876605  subbranch -> subbranch

Mourian P@MOURIANPARTHASARATHY MINGW64 ~/mourianp/mourianp (subbranch)
$ git status
On branch subbranch
Your branch is up to date with 'origin/subbranch'.

nothing to commit, working tree clean

Mourian P@MOURIANPARTHASARATHY MINGW64 ~/mourianp/mourianp (subbranch)
$ git pull origin subbranch2
From https://github.com/Mourian-sys/mourianp
 * branch            subbranch2 -> FETCH_HEAD
 * [new branch]      subbranch2 -> origin/subbranch2
Already up to date.

Mourian P@MOURIANPARTHASARATHY MINGW64 ~/mourianp/mourianp (subbranch)
$ git switch subbranch2
branch 'subbranch2' set up to track 'origin/subbranch2'.
Switched to a new branch 'subbranch2'

Mourian P@MOURIANPARTHASARATHY MINGW64 ~/mourianp/mourianp (subbranch2)
$ git pull origin subbranch
From https://github.com/Mourian-sys/mourianp
 * branch            subbranch  -> FETCH_HEAD
Already up to date.

Mourian P@MOURIANPARTHASARATHY MINGW64 ~/mourianp/mourianp (subbranch2)
$ git add subbranch
fatal: pathspec 'subbranch' did not match any files

Mourian P@MOURIANPARTHASARATHY MINGW64 ~/mourianp/mourianp (subbranch2)
$ git commit -m "new2"
On branch subbranch2
Your branch is up to date with 'origin/subbranch2'.

nothing to commit, working tree clean

Mourian P@MOURIANPARTHASARATHY MINGW64 ~/mourianp/mourianp (subbranch2)
$ ls -l
total 2
-rw-r--r-- 1 Mourian P 197121 10 Jan  5 18:45 README.md
-rw-r--r-- 1 Mourian P 197121 27 Jan  5 19:13 file.py

Mourian P@MOURIANPARTHASARATHY MINGW64 ~/mourianp/mourianp (subbranch2)
$ git push origin subbranch2
Everything up-to-date

Mourian P@MOURIANPARTHASARATHY MINGW64 ~/mourianp/mourianp (subbranch2)
$
```


AWS Management Console
Console Home | Console Home
git login - Yahoo India Search R
Mourian-sys/mourianp at subbr
FPS N/A | GPU 0% 38°C | CPU 13% | LAT N/A SM N/A
github.com/Mourian-sys/mourianp/tree/subbranch2
Mourian-sys / mourianp
Type / to search
Code Issues Pull requests Actions Projects Wiki Security Insights Settings
mourianp Public
Pin Watch 0 Fork 0 Star 0
Mourian-sys-patch-1 had recent pushes 33 minutes ago
Compare & pull request
subbranch2 had recent pushes 3 seconds ago
Compare & pull request
subbranch2 4 Branches 0 Tags Go to file t Add file Code
This branch is 2 commits ahead of main .
Contribute
Mourian-sys new f876605 · 26 minutes ago 3 Commits
README.md Initial commit 1 hour ago
file.py new 26 minutes ago
README
mourianp
About
No description, website, or topics provided.
Readme
Activity
0 stars
0 watching
0 forks
Releases
No releases published
Create a new release
Packages
No packages published
Publish your first package
25°C Partly cloudy
Search
ENG IN
19:50 05-01-2026


AWS Management Console
Console Home | Console Home
git login - Yahoo India Search R
Mourian-sys/mourianp at subbr
FPS N/A | GPU 0% 39°C | CPU 12% | LAT N/A SM N/A
github.com/Mourian-sys/mourianp/tree/subbranch
Mourian-sys / mourianp
Type / to search
Code Issues Pull requests Actions Projects Wiki Security Insights Settings
mourianp Public
Pin Watch 0 Fork 0 Star 0
Mourian-sys-patch-1 had recent pushes 54 minutes ago
Compare & pull request
subbranch2 had recent pushes 21 minutes ago
Compare & pull request
subbranch 4 Branches 0 Tags Go to file t Add file Code
This branch is 2 commits ahead of main .
Contribute
Mourian-sys new f876605 · 27 minutes ago 3 Commits
README.md Initial commit 1 hour ago
file.py new 27 minutes ago
README
mourianp
About
No description, website, or topics provided.
Readme
Activity
0 stars
0 watching
0 forks
Releases
No releases published
Create a new release
Packages
No packages published
Publish your first package
25°C Partly cloudy
Search
ENG IN
19:50 05-01-2026

HERE TWO BRANCHES ARE CREATED AND A FILE HAS CREATED IN A SUBBRANCH AND A FILE HAS PULL AND PUSH TO A SUBRANCH2 ,

---

1. git branch

- Used to **create, list, or delete branches**.
- Branch helps work on features without affecting main code.
- Example:
  git branch dev → creates a branch named *dev*

---

2. git add

- Used to **add files to staging area** before commit.
- Prepares changes to be saved.
- Example:
  git add file.txt
  git add . (adds all files)

---

3. git commit -m

- Saves staged changes with a **message**.
- Creates a snapshot of code.
- Example:
  git commit -m "Added login feature"

---

4. git remote

- Used to **connect local repo to remote repo** (GitHub).
- Shows remote repository URL.
- Example:
  git remote -v

---

5. git push origin

- Uploads local commits to **remote repository**.
- Shares code with others.
- Example:
  git push origin main

---

6. git pull

- Downloads and merges changes from remote to local.
- Keeps local repo updated.
- Example:
  git pull

---

7. git pull origin

- Pulls changes from a **specific remote**.
- Used when multiple remotes exist.
- Example:
  git pull origin main

---

8. git merge

- Combines one branch into another.
- Used after feature completion.
- Example:
  git merge dev

---

9. git rebase

- Moves commits to a new base branch.
- Keeps history **clean and linear**.
- Example:
  git rebase main